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SHAPING THE FUTURE OF CHEMISTRY LEARNING: THE IMPACT OF PLAYFUL STRATEGIES AND DIGITAL TOOLS

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ABSTRACT

Objective: This study examines the effectiveness of a methodological proposal that employs playful strategies through digital tools in the learning process of chemistry to improve the academic performance of first-semester students.

Theoretical Framework: This research is grounded in theories of playful learning, gamification, and educational technology, highlighting the role of interactive and immersive methodologies in enhancing student motivation, knowledge retention, and overall academic success.

Method: A quasi-experimental design with pre-test, post-test, and control group was employed to evaluate the effectiveness of the methodological proposal. The General Linear Model (GLM) for repeated measures was used for data analysis, with intra-subject factors (pre-test and post-test) and an inter-subject factor (experimental group vs. control group).

Results and Discussion: The implementation of playful strategies positively influenced students' academic performance in certain areas of the course. However, not all units showed significant improvement, suggesting that the impact of these methodologies may vary depending on the content and context of the learning material.
Research Implications: The study highlights the potential of playful strategies to promote not only better learning outcomes but also a more positive attitude toward education.

Originality/Value: This research demonstrates the adaptability and effectiveness of playful learning and gamification in higher education settings, offering valuable insights for designing immersive learning experiences tailored to students' specific needs.

Keywords: Chemistry Education, Educational Technology, Gamification, Playful Learning, Academic Engagement.

TRANSFORMANDO O APRENDIZADO DE QUÍMICA: O IMPACTO DAS ESTRATÉGIAS LÚDICAS E FERRAMENTAS DIGITAIS

RESUMO

Objetivo: Este estudo examina a eficácia de uma proposta metodológica que utiliza estratégias lúdicas por meio de ferramentas digitais no processo de aprendizagem de química para melhorar o desempenho acadêmico dos estudantes do primeiro semestre.

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Marco Teórico: Esta pesquisa baseia-se em teorias de aprendizagem lúdica, gamificação e tecnologia educacional, destacando o papel de metodologias interativas e imersivas em aumentar a motivação dos estudantes, a retenção do conhecimento e o sucesso acadêmico geral.

Método: Foi utilizado um desenho quase-experimental com pré-teste, pós-teste e grupo de controle para avaliar a eficácia da proposta metodológica. O Modelo Linear Geral (MLG) para medidas repetidas foi empregado na análise dos dados, com fatores intra-sujeitos (pré-teste e pós-teste) e um fator inter-sujeitos (grupo experimental vs. grupo de controle).

Resultados e Discussão: A implementação de estratégias lúdicas influenciou positivamente o desempenho acadêmico dos estudantes em algumas áreas do curso. No entanto, nem todas as unidades apresentaram melhora significativa, sugerindo que o impacto dessas metodologias pode variar dependendo do conteúdo e do contexto do material de aprendizagem.

Implicações da Pesquisa: O estudo destaca o potencial das estratégias lúdicas para promover não apenas melhores resultados de aprendizagem, mas também uma atitude mais positiva em relação à educação.

Originalidade/Valor: Esta pesquisa demonstra a adaptabilidade e a eficácia da aprendizagem lúdica e da gamificação em ambientes de ensino superior, oferecendo insights valiosos para projetar experiências de aprendizagem imersivas que atendam às necessidades específicas dos estudantes.

Palavras-chave: Educação em Química, Tecnologia Educacional, Gamificação, Aprendizagem Lúdica, Compromisso Acadêmico.

MOLDEANDO EL FUTURO DEL APRENDIZAJE DE LA QUÍMICA: EL IMPACTO DE LAS ESTRATEGIAS LÚDICAS Y LAS HERRAMIENTAS DIGITALES

RESUMEN

Objetivo: Este estudio examina la efectividad de una propuesta metodológica que emplea las estrategias lúdicas a través de herramientas digitales en el proceso de aprendizaje de la químical para la mejora del rendimiento académico de los estudiantes de primer semestre.

Marco Teórico: Esta investigación se basa en teorías de aprendizaje lúdico, gamificación y tecnología educativa, destacando el papel de las metodologías interactivas e inmersivas en aumentar la motivación de los estudiantes, la retención del conocimiento y el éxito académico general.

Método: Se empleó un diseño cuasiexperimental con pretest, postest y grupo control para evaluar la efectividad de la propuesta metodológica. Para el análisis de datos se empeló el Modelo Lineal General (MLG) de medidas repetidas, con factores intra-sujetos (pretest y postest) y factor inter-sujetos (grupo experimental vs. grupo control).

Resultados y Discusión: La implementación de estrategias lúdicas influyó positivamente en el rendimiento académico de los estudiantes en algunas áreas del curso. Sin embargo, no todas las unidades mostraron una mejora significativa, lo que sugiere que el impacto de estas metodologías puede variar dependiendo del contenido y del contexto del material de aprendizaje.

Implicaciones de la Investigación: Se destaca el potencial de las estrategias lúdicas para promover no solo mejores resultados de aprendizaje, sino también una actitud más positiva hacia la educación.

Originalidad/Valor: Esta investigación demuestra la adaptabilidad y la efectividad del aprendizaje lúdico y la gamificación en entornos de educación superior, ofreciendo valiosos conocimientos para diseñar experiencias de aprendizaje inmersivas que respondan a las necesidades específicas de los estudiantes.

Palabras clave: Educación en Química, Tecnología Educativa, Gamificación, Aprendizaje Lúdico, Compromiso Académico.

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1 INTRODUCTION

In recent years, educational innovation has been increasingly driven by the integration of digital tools and playful strategies, particularly in higher education. This shift reflects the need to adapt teaching methodologies to the preferences and learning styles of a new generation of students, characterized by their familiarity with technology and preference for interactive, engaging experiences. Chemistry education, especially in foundational courses like General Chemistry I, often presents unique challenges, including abstract concepts, complex problem-solving requirements, and the need to relate theoretical knowledge to practical applications. These challenges frequently result in low student motivation, high dropout rates, and suboptimal academic performance.

The potential of playful learning and gamification to address these challenges has been well-documented in the literature. Playful learning leverages enjoyment, interaction, and curiosity to foster engagement, while gamification applies game design elements—such as rewards, competition, and progression mechanics—to non-game contexts, including education (Calderón *et al.*, 2024; Czakóová, 2019). Together, these approaches provide a framework for creating immersive, motivational learning environments that not only enhance cognitive outcomes but also promote positive emotional and attitudinal changes in learners.

Mobile devices, which are integral to students' daily lives, offer a versatile platform for implementing playful learning strategies that promote dynamic and personalized learning experiences. Their accessibility and interactivity provide an opportunity to meet the evolving demands of higher education. This study examines whether playful strategies, delivered through mobile devices, can significantly improve academic performance in General Chemistry I, a course known for its abstract concepts and complex problem-solving, which many STEM students find challenging.

Although research on the use of playful learning in chemistry remains limited, this study aims to fill two key gaps: first, the effectiveness of playful strategies in chemistry education, and second, the role of mobile devices in supporting these strategies. By exploring these aspects, the research seeks to offer actionable insights into enhancing student outcomes in STEM disciplines.

The general objective of this study is to analyze whether the use of playful strategies via mobile devices positively impacts the academic performance of students in General Chemistry I. Specifically, the study aims to design, implement, and evaluate a didactic intervention for chemistry learning based on playful strategies using mobile devices. Additionally, the study

will assess the effectiveness of these playful strategies in improving students' academic performance in the course.

The general hypothesis posits that the use of playful strategies through mobile devices will significantly enhance students' academic performance in General Chemistry I. More specifically, it is hypothesized that a didactic intervention incorporating playful strategies and mobile devices can be successfully designed, implemented, and evaluated. Furthermore, it is expected that students in the experimental group will show a significant improvement in their academic performance compared to those in the control group.

Through these objectives and hypotheses, the study aims to provide empirical evidence on how innovative teaching methods can address the challenges faced in STEM education. The findings will contribute to educational innovation, offering valuable insights and resources for designing effective teaching interventions in higher education.

2 THEORETICAL FRAMEWORK

The theoretical framework in a study comprises a critical and organized analysis of the literature relevant to the topic, providing a theoretical contextualization and defining the key concepts. It must comprehensively contain theories, models and previous research, identifying gaps, contradictions and consensuses in the literature that are important for the focus of the work being developed.

The use of playful strategies and digital tools in education is supported by various theories and models that emphasize the importance of engagement, motivation, and interaction in the learning process (Ballatore & Tabacco, 2024; Playfoot, 2016; Playfoot *et al.*, 2017). This section provides a critical and organized review of the literature, highlighting the theoretical foundations, relevant models, and previous research that underpin the focus of this study.

2.1 PLAYFUL LEARNING AND GAMIFICATION IN EDUCATION

Playful learning is grounded in constructivist and experiential learning theories (Ahmed & Sutton, 2017; Despeisse, 2018; Tsang *et al.*, 2024; Usman *et al.*, 2023) which emphasize the active role of learners in constructing knowledge through meaningful and engaging experiences. Playful learning incorporates game-like elements into educational contexts, promoting curiosity, exploration, and enjoyment as central components of the learning process. It aligns with recent literature research (Ilbeigi *et al.*, 2024; Massad, 2024; Putra & Suhaimi,

2024; Zhang, 2024), which underscores the importance of interaction and social dynamics in fostering cognitive development.

Gamification, a specific application of playful learning, refers to the use of game design elements—such as points, badges, leaderboards, and challenges—in non-game contexts, including education (Alcaraz *et al.*, 2019; Behl *et al.*, 2024; Salleh *et al.*, 2024; Stadnicka & Deif, 2019). This approach seeks to harness the motivational aspects of games to increase student engagement and participation. Gamification positively influences learning outcomes by enhancing motivation, satisfaction, and performance, particularly in STEM disciplines (Flores-Nicolás & Reyes, 2022; Forndran & Zacharias, 2019; Halvorsen *et al.*, 2019; Indriasari *et al.*, 2020).

2.2 MOBILE DEVICES AND DIGITAL TOOLS IN EDUCATION

The integration of mobile devices into educational practices is supported by the theory of mobile learning (m-learning), which emphasizes the flexibility, accessibility, and contextual relevance of mobile technology. Mobile devices enable personalized, on-the-go learning experiences, fostering autonomy and adaptability in students (Candel & Colmenero, 2022; Choi-Lundberg *et al.*, 2023; Maev & Zhiltsov, 2021; Martínez *et al.*, 2016). It highlights that m-learning enhances interaction, collaboration, and real-time feedback, making it a valuable tool for addressing diverse learning needs.

Previous research has demonstrated the effectiveness of mobile-based learning in improving academic performance across various disciplines. In this context, the scientific literature underscores the potential of mobile devices to transform traditional educational practices, particularly when integrated with gamified and playful strategies (Elsherbiny & Al Maamari, 2021; Jacobs *et al.*, 2023; Lytvynov *et al.*, 2022; Terzieva, 2019).

2.3 CHEMISTRY EDUCATION AND INNOVATIVE METHODOLOGIES

Chemistry education often presents unique challenges, including abstract concepts, complex problem-solving, and the need for practical application of theoretical knowledge. Traditional teaching methods, such as lectures and rote memorization, are often insufficient to address these challenges (Alberto *et al.*, 2024; Alwassil *et al.*, 2024; Marín *et al.*, 2020). Instead, innovative methodologies that incorporate interactive and immersive elements have been shown to enhance student engagement and understanding.

Playful and gamified strategies have gained prominence in chemistry education as effective tools for improving student motivation and learning outcomes (Guerrero *et al.*, 2012; Hanson *et al.*, 2014; Molina & Fernandez, 2010; Sultanova *et al.*, 2021). These approaches leverage the engaging and interactive nature of games to address the challenges often associated with teaching abstract and complex chemistry concepts.

Research indicates that gamification in chemistry classrooms fosters a deeper understanding of theoretical and practical aspects by creating a dynamic and participatory learning environment (Janstová *et al.*, 2013; Miranda *et al.*, 2023; Ramos & Condotta, 2024). For instance, incorporating game elements such as challenges, rewards, and storytelling has been shown to enhance students' ability to grasp difficult topics. Similarly, game-based learning environments, which immerse students in problem-solving scenarios, not only improve critical thinking skills but also help them develop strategies for real-world applications of chemistry (López *et al.*, 2015; Monclús *et al.*, 2019; Sugano & Nabua, 2020). Such methodologies also promote collaboration and communication among peers, further enriching the learning experience. These findings highlight the transformative potential of playful and gamified strategies in making chemistry education more effective and engaging for students.

2.4 GAPS, CONTRADICTIONS, AND CONSENSUSES

Despite the growing body of literature supporting playful and gamified learning, several gaps and contradictions remain. While many studies report positive outcomes, others suggest that the effectiveness of these methodologies may vary depending on factors such as the subject matter, student demographics, and implementation quality (Leou *et al.*, 2006; Martín-Fernández *et al.*, 2016; Pizarro, 2018; Wong *et al.*, 2021). Additionally, there is limited research on the application of these strategies in chemistry education, particularly in higher education contexts.

A consensus exists on the potential of digital tools and playful strategies to enhance motivation and engagement. However, there is a need for more rigorous, context-specific studies to examine their impact on academic performance, particularly in STEM fields (Cardenas *et al.*, 2022; Hou, 2013; Kovacs *et al.*, 2019a; Morton & Colbert-Getz, 2017). This research aims to address these gaps by exploring the integration of playful strategies and mobile devices in General Chemistry I courses, contributing to the growing body of knowledge on innovative teaching methodologies.

In summary, the theoretical framework of this study is rooted in constructivist learning

theories, gamification, and mobile learning. By leveraging these theoretical foundations, this research seeks to provide empirical evidence on the effectiveness of playful and technology-enhanced learning in improving academic performance in chemistry education.

3 METHODOLOGY

3.1 PARTICIPANTS

This study used a quasi-experimental design to assess the impact of playful strategies combined with digital tools on the academic performance of General Chemistry I students. Conducted during the 2022-2023 academic year at the Central University of Ecuador, the sample consisted of 89 first-semester students from two sections (A and B). Section B (experimental group) and Section A (control group) were randomly selected. The majority of participants were female (65.2%), with males (33.7%) and a small proportion identifying as gender-diverse (1.1%). The average age was 19.7 years (SD = 1.9). This distribution ensured balanced conditions for evaluating the intervention's effectiveness.

3.2 MEASURES

The methodology section outlines the procedures undertaken in this study, ensuring transparency, replicability, and reliability. This includes the type of study, sample selection, methods of data collection and analysis, ethical considerations, and limitations. Below, the measures applied for data collection and intervention design are described in detail.

The instruments used in this study were selected and developed based on their relevance. To ensure the rigor of the process, all tools underwent expert evaluation and pilot testing with students at the same educational level, focusing on clarity and understanding.

Two instruments were employed:

To measure performance in the four units of the General Chemistry I course, two concept questionnaires were used: the pretest and the posttest. Both instruments consisted of 90 items organized according to the course units, assessing theoretical knowledge, technical vocabulary management, and the application of chemical concepts. The pretest evaluated initial knowledge before taking the course, using single-choice, multiple-choice, matching, completion, and true/false questions, while the posttest, applied after completing the course, maintained the same format and included additional questions on sequencing and short-answer

responses, focusing on the ability to apply acquired knowledge.

The intervention aimed to improve academic performance in General Chemistry I by integrating playful activities, gamification, and mobile device use. The intervention followed a structured timeline, implemented in three stages: initial (consent and baseline), implementation (application of strategies), and evaluation (posttest and outcome assessment).

Table 1

Implementation Timeline and Key Activities for Content Development and Evaluation

Month	Activity
May	Introduction to the project, informed consent, pretest administration, and Unit I content development.
June	Application of playful strategies for Unit I and Unit I evaluation. Development of Unit II content.
July	Application of playful strategies for Unit II and Unit II evaluation. Development of Unit III content.
August	Playful strategies for Unit III, Unit III evaluation, gamification for Unit IV, Unit IV evaluation, and posttest.

3.4 PROCEDURE

In the first stage, the experimental group was randomly selected, and informed consent was obtained. Pre-questionnaires on playful strategies, internet usage, and chemistry concepts were administered to both groups. Playful strategies were designed in collaboration with students, tailored to their characteristics, and aligned with course objectives. In the second stage, playful activities were implemented during the reinforcement phase, focusing on reviewing and consolidating content before assessments, with support from available equipment and materials. The activities were structured in three phases: the first focused on individual work to reinforce concepts; the second promoted collaborative learning through group discussions and problem-solving; and the third involved a gamified tournament to apply learning in a competitive, dynamic setting, using digital tools that provided real-time feedback.

3.5 DATA DESIGN AND ANALYSIS

To assess the effectiveness of the methodological proposal integrated into the chemistry learning process of students enrolled in the General Chemistry I course, a quasi-experimental design with pretest and posttest measurements and a control group was employed. The data analysis was carried out using the General Linear Model (GLM) for repeated measures. For this, the pretest and posttest scores of the evaluated variables were considered as within-subjects

factors, while the between-subjects factor was the group assignment, either control or experimental.

The data analysis and graphical representation were conducted using the SPSS v.22 software, licensed to the University of Alicante. This approach allowed for a detailed comparison of the changes in the students' learning outcomes and the evaluation of the impact of the proposed gamified methodology on their understanding of chemistry.

4 RESULTS AND DISCUSSIONS

The objectives set for this research have been achieved, as evidenced by the significant improvement in the academic performance of the experimental group following the methodological intervention. Participants in the experimental group demonstrated notable progress in their performance in General Chemistry I, as reflected in the scores obtained on the posttest knowledge (please, see table 2).

The improvement was significant in two of the four units of the course: Unit III, which focuses on the periodic table, and Unit IV, which covers inorganic chemical nomenclature. Among these, Unit IV stood out for its complexity, as previously noted by students from earlier cohorts. These results underscore the importance of incorporating playful strategies into educational practices, demonstrating their effectiveness in achieving both pedagogical and academic objectives. This was evident even in units traditionally considered more challenging for students.

The quantitative analysis of academic performance provided an objective evaluation of the program's effectiveness, supporting its value as a formal educational approach. Additionally, the findings highlight the potential of rigorous pedagogical interventions when working with adult learners in higher education settings. By expanding the application of playful strategies to a wider age range, this research emphasizes the intrinsic value of playful activities as a universal human quality that can enhance learning at any stage of life.

Table 2

Summary of intra-inter subject univariate analysis of variance

	Source	Type III	df	F	Sig.	η2 partial	Ob. Power
Academic Performance in the Subject	Intra	23.415.062	1	443.536	.000**	.836	1.000
	Intra*Entre	1.088.522	1	20.619	.000**	.192	.994
	Error intra	4.592.882	87				
	Inter	311.946.687	1	1.966.400	.000**	.958	1.000
	Condition	667.698	1	4.209	.043*	.046	.528
	Error inter	13.801.549	87				
Academic Performance in Unit I	Intra	462.433	1	105.261	.000**	.547	1.000
	Intra*Entre	3.264	1	.743	.391	.008	.137
	Error Intra	382.207	87				
	Inter	13.554.063	1	1.379.359	.000**	.941	1000
	Condition	9.231	1	.939	.335	.011	.160
	Error inter	854.892	87				
Academic Performance in Unit II	Intra	1.735.605	1	303.276	.000**	.777	1
	Intra*Entre	5.133	1	.897	.346	.010	.155
	Error Intra	497.889	87				
	Inter	16.890.952	1	1.956.879	.000**	.957	1.000
	Condition	.165	1	.019	.890	.000	.052
	Error inter	750.947	87				
Academic Performance in Unit III	Intra	1.421.634	1	240.451	.000**	.734	1.000
	Intra*Entre	138.578	1	23.439	.000**	.212	.998
	Error Intra	514.377	87				
	Inter	18.067.984	1	1.220.980	.000**	.933	1.000
	Condition	91.624	1	6.192	.015*	.066	.692
	Error inter	1.287.420	87				
Academic Performance in Unit IV	Intra	2.535.328	1	138.810	.000**	.615	1.000
	Intra*Entre	248.856	1	13.625	.000**	.135	.955
	Error Intra	1.589.031	87				
	Inter	30.947.571	1	1.128.814	.000**	.928	1.000
	Condition	336.335	1	12.268	.001**	.124	.934
	Error inter	2.385.193	87				

*p < .05, **p<.01.

The findings align with recent scientific literature, suggesting that with careful planning and effective implementation, educational programs incorporating playful strategies and gamification can yield positive learning outcomes (Alwassil *et al.*, 2024; Lin *et al.*, 2024; Ramos & Condotta, 2024; Sultanova *et al.*, 2021). The alignment between the program's implementation and its educational objectives, as well as the adherence to gamification principles, was crucial for the achievement of the set goals.

A clear connection was observed between the learning results and the strength of the playful strategies used. Adjustments made to the strategies in later units, based on feedback from earlier ones, played a significant role in meeting course objectives. These results highlight the importance of a structured and reflective approach when integrating playful activities into education, emphasizing the need for careful planning, precision, and a clear distinction between gamification and other game-based methodologies.

The planning and implementation of playful strategies is an ongoing learning process that improves with experience and reflection. While the first two units showed limited improvements in academic performance, the efforts made during this phase were still valuable in shaping the overall teaching and learning dynamics. This indicates that even early challenges contribute to the learning process, laying the groundwork for the further development of playful teaching methods, in line with other research (Alvarez-Corral *et al.*, 2010; de Souza *et al.*, 2019; Leou *et al.*, 2006; Martín-Fernández *et al.*, 2016; Pizarro, 2018). Continuous reflection and feedback from participants provide crucial insights for refining and enhancing future implementations.

The intervention program also highlights the potential to redefine the learning process itself as an innate human quality that develops naturally through curiosity and exploration. By creating comfortable and engaging environments, the program encourages interaction and collaboration among learners, fostering values such as empathy, solidarity, and friendship. These are just a few of the many benefits that playful learning offers, enhancing both academic and personal growth, in line with other studies (Hanson *et al.*, 2014; Jaramillo *et al.*, 2024; Perminov *et al.*, 2019; Wong *et al.*, 2021).

The program's focus on play and games fosters key cognitive processes such as attention, memory, and creativity, leading to richer, more meaningful learning experiences. It also helps students overcome common obstacles such as anxiety, mental blocks, and feelings of failure, which can impede academic success. Moreover, playful strategies can address broader social challenges, including violence and school disengagement, by offering students constructive outlets for negative emotions, thereby improving their overall performance.

Incorporating ICT tools in the classroom caters to the digital preferences and needs of today's students (Castro & Andrade, 2011; Crivellaro *et al.*, 2015; Pizarro, 2018; Yusuf *et al.*, 2017). This not only makes the learning experience more engaging and relevant but also allows for greater flexibility in teaching. By removing barriers related to time, space, and geography, these tools support personalized and inclusive learning.

The use of Interactive Groups further strengthens the inclusivity of the program, promoting meaningful learning through interaction in equal and diverse settings. This approach enhances cognitive development and values such as tolerance, fostering deeper engagement among students and other educational stakeholders.

Teacher involvement in playful activities nurtures positive relationships with students, as these activities encourage spontaneous expressions of joy and empathy. This connection enhances students' self-esteem, social skills, and academic performance (Kovacs *et al.*, 2019a;

Sugano & Nabua, 2020; Sultanova *et al.*, 2021; Yoel & Dori, 2022). It also allows teachers to better understand their students' academic and emotional needs, which may not always be visible in conventional classes.

Collaborating with students to design playful strategies also increases creativity and broadens the use of digital tools, helping bridge generational gaps and enriching the learning process. This collaborative effort benefits both students and teachers, enhancing their digital competencies and fostering a mutually supportive learning environment (de Souza *et al.*, 2019; Monclús *et al.*, 2019; Obada *et al.*, 2023; Ortuzar *et al.*, 2019).

The thoughtful design of playful activities ensures they align with the desired educational outcomes, preventing the inclusion of activities that may be entertaining but lack educational value. Each activity is carefully chosen to serve a pedagogical purpose, contributing to the emotional, intellectual, and social development of students (Alvarez-Corral *et al.*, 2010; Jaña *et al.*, 2015; Kovacs *et al.*, 2019b; Morton & Colbert-Getz, 2017).

Moreover, the program emphasizes intrinsic motivation, recognizing its essential role in fostering meaningful and satisfying learning experiences. While extrinsic motivation is considered, the primary focus is on motivating students to engage with the content in a way that supports both their academic and personal growth.

However, the intervention program had some limitations, including limited impact on the first two units due to the short implementation time and initial design of the activities. Despite this, the groundwork provides valuable insights for future interventions. The reliance on digital tools also posed challenges in schools with limited resources, although their potential for engagement was evident. The small sample size restricted the generalizability of the findings, and future studies should include larger, diverse samples to better assess scalability. Other factors, such as student motivation, were not considered and should be addressed in future research. Additionally, the limited application of playful strategies, primarily in the reinforcement phase, may have hindered their full potential; extending these strategies throughout the entire course could enhance the learning experience. Despite these limitations, the program showed promise, and future research will refine these approaches for greater impact.

5 CONCLUSION

In conclusion, the proposed program demonstrated positive effects on students' academic performance in General Chemistry I, particularly in enhancing the learning process,

despite not showing significant results across all units. The integration of playful strategies into higher education underscores the value of play in fostering engagement and learning, confirming that these approaches are beneficial across all educational levels and for individuals of various ages. The versatility of gamification, when combined with other innovative teaching methods, emerged as an effective tool for improving student engagement and learning outcomes. However, the success of these strategies hinges on the dedication, expertise, and adaptability of educators, who play a pivotal role in designing and leading activities aligned with educational objectives. This study highlights the importance of further expanding playful strategies across the curriculum to enhance their effectiveness, suggesting significant potential for future research and application.

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